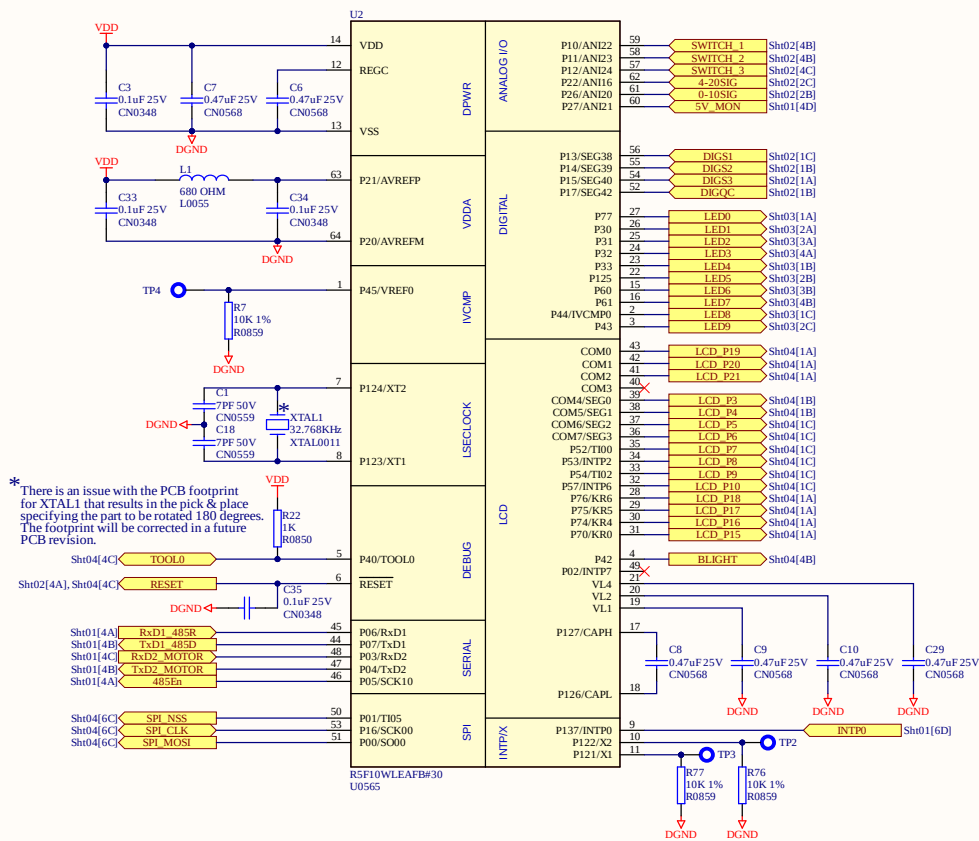
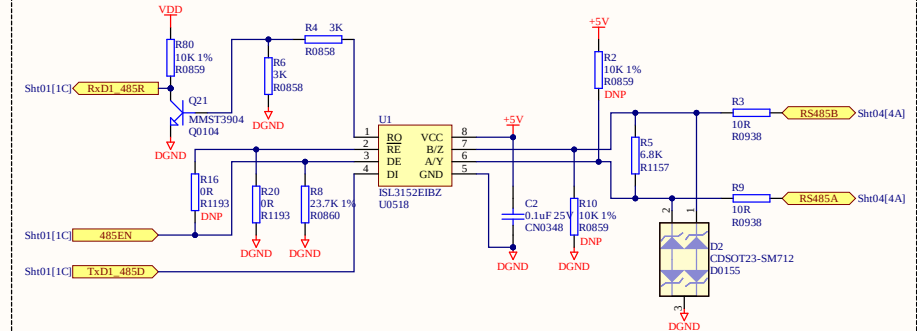


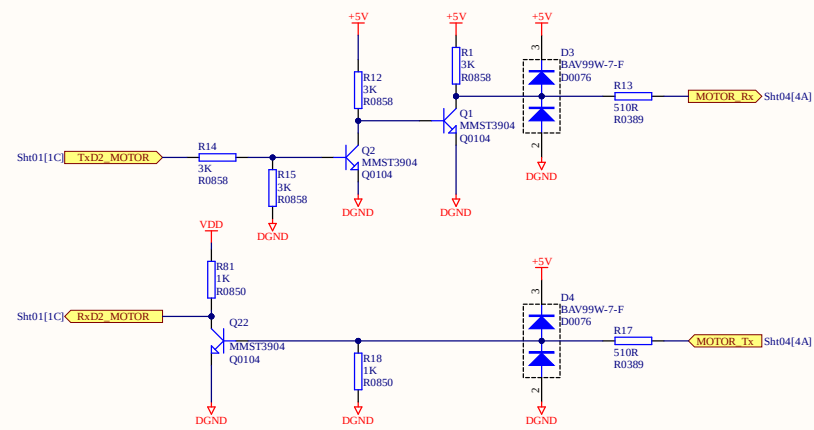
NIDEC DRIVE SYSTEMS
KCSC0231 Rev G
5/23/2024



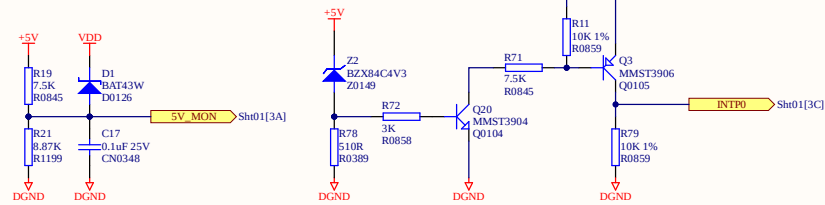
RS485 COMMUNICATION



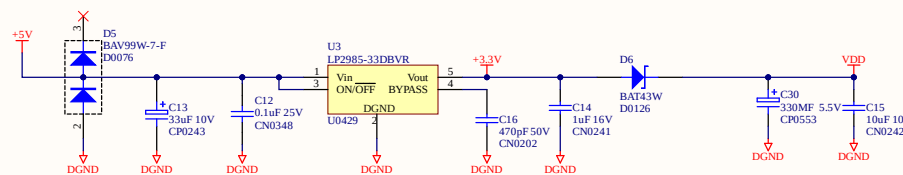
MOTOR COMMUNICATION



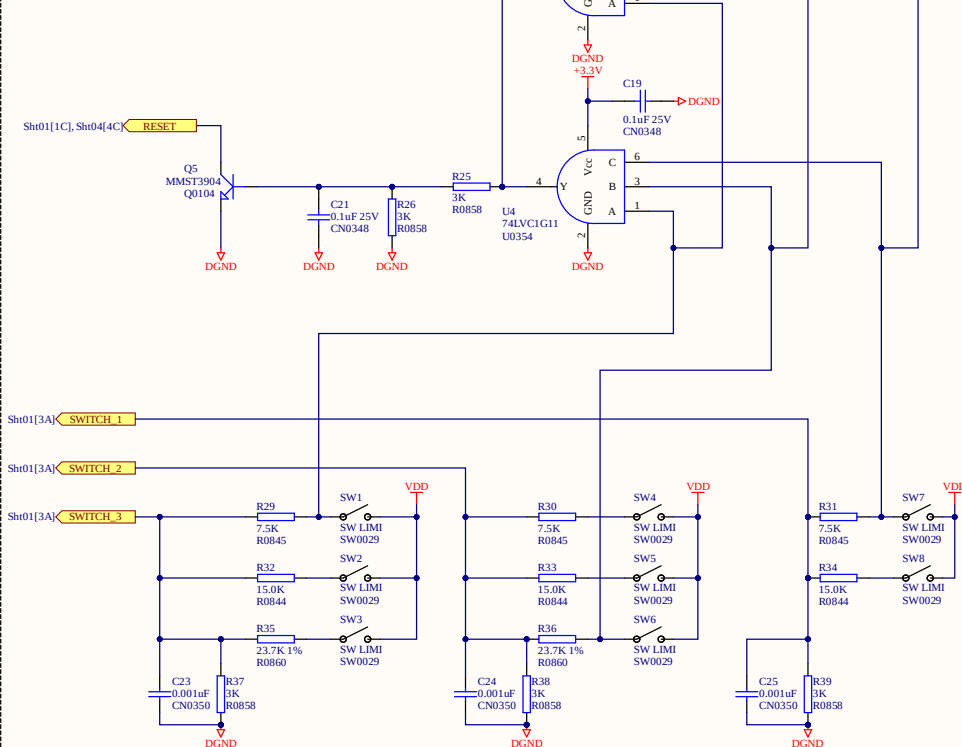
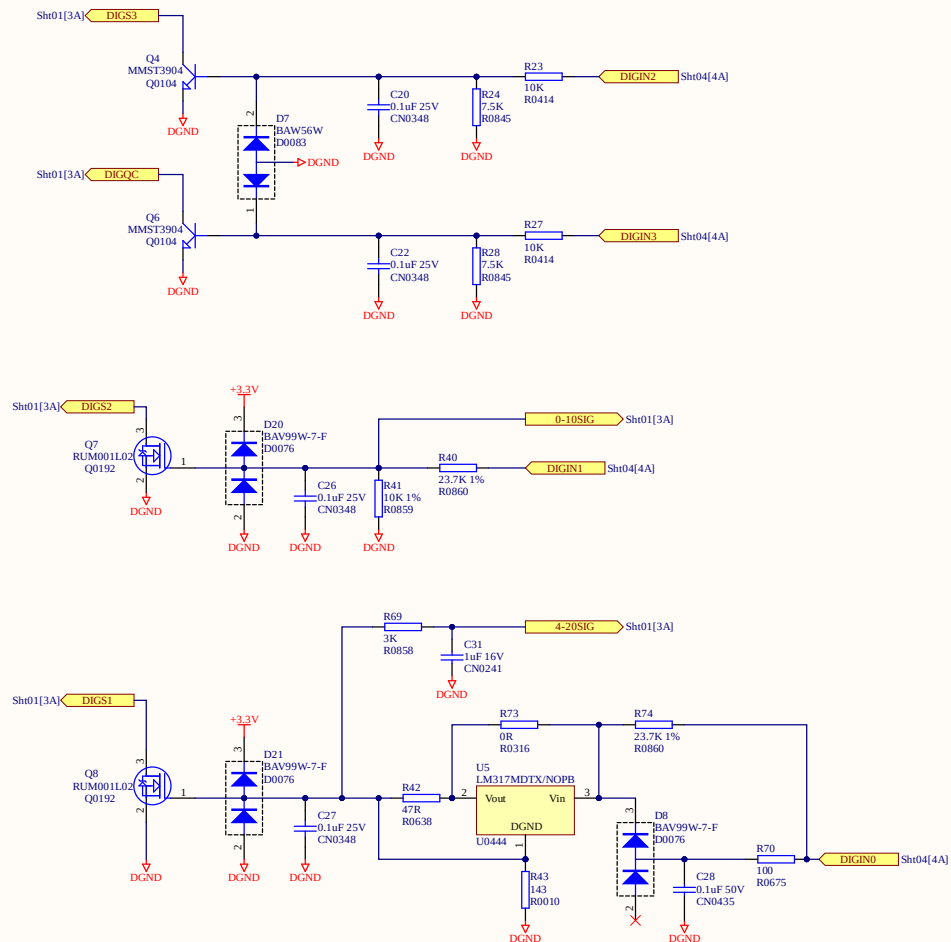
+5V MONITOR



POWER SUPPLY



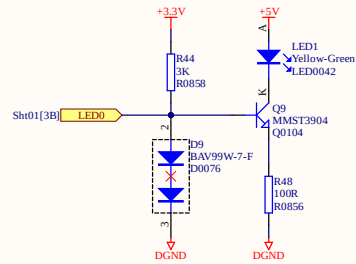
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HMI 3HP Roadrunner			
Size	Number	Revision	
Tabloid	KCSC0231	G	
Date: 5/23/2024		Sheet: 1 of 5	
File: C:\Users\Sh01\SCHDOC		Drawn By: Kurt Erickson	



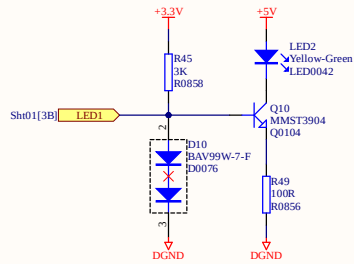
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HMI 3HP Roadrunner			
Size	Number	Revision	
Tabloid	KCSC0231	G	
Date: 5/23/2024		Sheet:	2 of 5
File: C:\Users\...\Sht02.SCHDOC		Drawn By:	Kurt Erickson

LED DRIVE

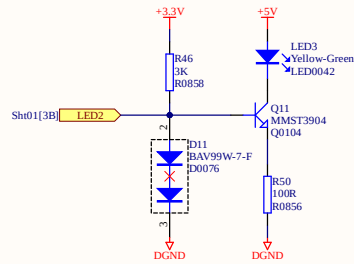
SPEED



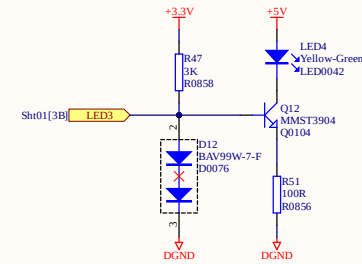
TIME



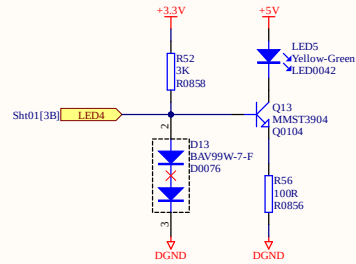
DURATION



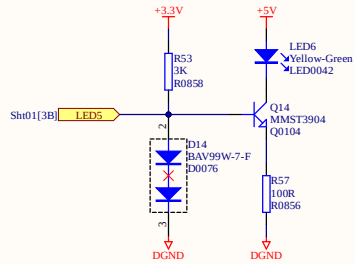
WATTS



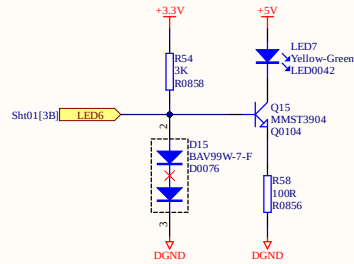
S1



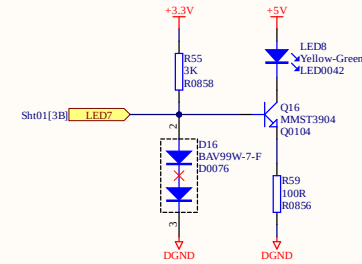
S2



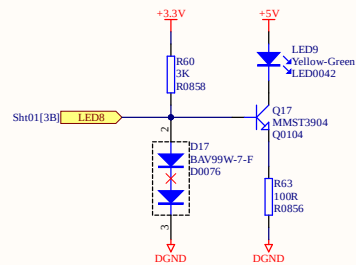
S3



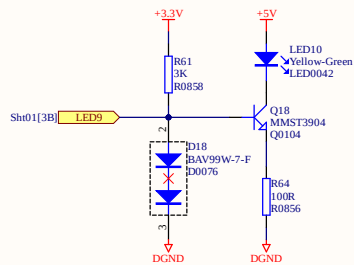
QUICK CLEAN



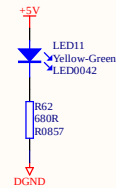
START/STOP



AUTOMATION ONLY



LIGHTNING BOLT



Title			
HMI 3HP Roadrunner			
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Diagram illustrating the connections for LCD01 (Orient Display, AM0463R-03, LCD0014) to the AM0463R-03 LCD0014 module.

The connections are as follows:

- Sh01[3B] LCD_P15 connects to pin 23.
- Sh01[3B] LCD_P16 connects to pin 22.
- Sh01[3B] LCD_P17 connects to pin 21.
- Sh01[3B] LCD_P18 connects to pin 20.
- Sh01[3B] LCD_P19 connects to pin 19.
- Sh01[3B] LCD_P20 connects to pin 18.
- Sh01[3B] LCD_P21 connects to pin 17.
- Sh01[3B] LCD_P3 connects to pin 1.
- Sh01[3B] LCD_P4 connects to pin 2.
- Sh01[3B] LCD_P5 connects to pin 3.
- Sh01[3B] LCD_P6 connects to pin 4.
- Sh01[3B] LCD_P7 connects to pin 5.
- Sh01[3B] LCD_P8 connects to pin 6.
- Sh01[3B] LCD_P9 connects to pin 7.
- Sh01[3B] LCD_P10 connects to pin 8.

The module is labeled AM0463R-03 LCD0014.

LCD1
Orient Display,AM0463R-03
LCD0014

[illegible]

Comment
J0342

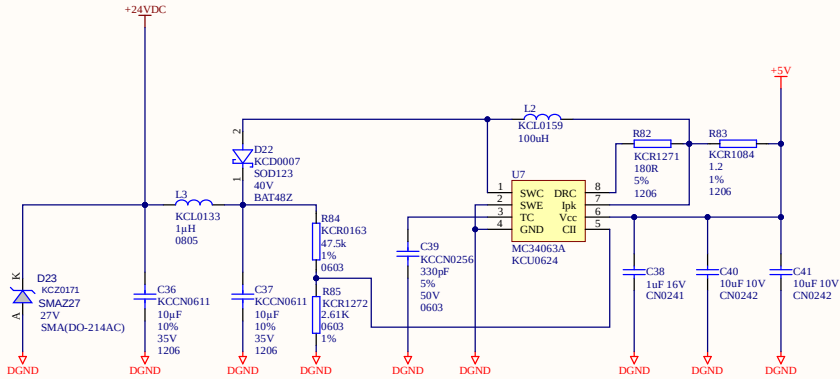
The diagram illustrates the electrical connection for the LCD LED backlight. It features a power supply section with a +5V rail and a DGND (Ground) rail. A resistor network, consisting of R66 (510R) and R0389, is connected to the +5V rail. A transistor, Q19 (MMST3904), is used to drive the backlight. The backlight is connected to a 3-pin connector (P3) labeled 'LCD LED BACKLIGHT (ALTERNATE CONNECTION)'. The connector pins are labeled A, K, and P. The diagram also includes labels for 'PADS FOR DIRECT BACKLIGHT SOLDERING' and a comment 'J0343'.

Pin connection diagram for the Molex 1051330001 J0423 DNP connector. The diagram shows a 6-pin connector with pins 1 through 6. Pin 1 is connected to VDD, pin 2 to Sht01[1C] SPI_CLK, pin 3 to Sht01[1C] SPI_NSS, pin 4 is marked with a red X, pin 5 to Sht01[1C] SPI_MOSI, and pin 6 to DGND. The connector is labeled J1, Tab, MOLEX, 1051330001, J0423, and DNP.

Title				HMI 3HP Roadrunner			
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File: C:\Users\...\Sht04.SCHDOC				Drawn By: Kurt Erickson			

+5V to +24V 100kHz BOOST CONVERTER SUPPLY

+24V OUT, 30mA MAX



This supply is used to power an external flow sensor on configuration G02.

REVISION LOG

REV A - ECN01536 - 08Mar2022
- Initial Release

REV B - ECN01544 - 21Mar2022
- Change U1 From R5F10WLCAFA#30 to R5F10WLEAFB#30

REV C - ECN01570 - 08Oct2022
- ADD D1
- Change R21 From 7.5K to 8.87K
- Change Pin of LED8 from P42(Pin4) to P44(Pin2)
- Change Pin of BLIGHT1 from P02(Pin49) to P42(Pin4)
- Change U2 pin 9 to "INTP0"
- ADD R80, R81, Q21, Q22.

REV D - ECN01741 - 15May2023
- Added +5V to +24V Boost Supply
- R81 was 10k

REV E - ECN01826 - 25Jan2024
- Moved configuration table Tab#1 from page 2 to page 5
- Added configuration G02 to configuration table
- Added note to XTAL1 on page 1

REV F - ECN01835 - 11Apr2024
- Added component info. to configuration table Tab#1
- Added LCD window dimension information to KCB0231 drawing

REV G - ECN01864 - 23May2024
- Change R18 from 3K to 1K
- Change R81 from 3K to 1K

TAB#1

	R43	R70	R73	R74	D8	D21	C28	U5	C32	R75	Z1	C36	C37	C38	C39	C40	C41	R65	R82	R83	R84	R85	R86	R87	D22	D23	L2	L3	U7
G00	1206 R0010 14J	1206 R0675 100	DNP	DNP	SOT323 D0076 BAV99W	DNP	0402 CN0435 0.1uF 50V	DPAK U0444 LM317	DNP	DNP	DNP	DNP	DNP	DNP	DNP	DNP	DNP	1206 R0817 240	DNP	DNP	DNP	DNP	DNP	R0316 0	DNP	DNP	DNP	DNP	DNP
G01	1206 R0765 10K	DNP	0603 R0316 0	0402 R0860 23.7K	DNP	SOT323 D0076 BAV99W	DNP	DNP	1210 CN0315 10nF 500V	1206 R0972 1M	SMD Z0142 GDT 600V	DNP	DNP	DNP	DNP	DNP	DNP	1206 R0817 240	DNP	DNP	DNP	DNP	DNP	R0316 0	DNP	DNP	DNP	DNP	DNP
G02	1206 R0010 14J	1206 R0675 100	DNP	DNP	SOT323 D0076 BAV99W	DNP	0402 CN0435 0.1uF 50V	DPAK U0444 LM317	1210 CN0315 10nF 500V	1206 R0972 1M	SMD Z0142 GDT 600V	1206 CN0611 10uF 35V	1206 CN0611 10uF 35V	0805 CN0241 1uF 16V	0603 CN0256 330pF 50V	0805 CN0242 10uF 10V	0805 CN0242 10uF 10V	1206 R0366	1206 R1271	1206 R1084	0803 R0163	0603 R1272	0603 R0316 0	DNP	SOD123 D0007 BAT48Z	SMA Z0171 SMAZZ7	SMD L0159 100uH	0805 L0133 1uH	SOP-8 U0624

Title			
HMI 3HP Roadrunner			
Size	Number	Revision	
Tabloid	KCSC0231	G	
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File: C:\Users\...\Sh05.SCHDOC		Drawn By:	Kurt Erickson